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Named entity recognition method in health preserving field based on BERT

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Abstract

With the aging of the population development, people pay more attention to health preserving. In order to build a knowledge graph of health-preserving field, named entity recognition is required first. In the health-preserving field, due to there is no publicly available data sets, building corpus by getting data from websites and defining seven types of entities. Considering the complexity and ambiguity of data, a named entity recognition method based on BERT in the health-preserving field is proposed. Because BERT can generate different vectors for the same word and make full use of context semantic. CNN can extract local features and BILSTM can capture long distance characteristics. Finally selecting the best sentence through the conditional random field. By comparing with other models in the same data set, it is verified that the new model is better in precision, recall and F_1 score, reaching the optimal value of 87.84%, which can meet the task requirements in the health-preserving field.

Keywords: BERT; health-preserving field; named entity recognition; deep learning

1. Overview

Nature Language Processing (NLP) is a hot research direction. It has important applications in machine translation, speech recognition, text classification, emotion analysis, question-answering system, knowledge graph,

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chatbot and other aspects. As a basic task of NLP, Name Entity Recognition (NER) aims to identify entities with specific meanings and types from unstructured text[1].

The method of named entity recognition has experienced continuous development, and some methods such as rule-based and dictionary-based, statistical machine learning and deep learning have been put forward. Among them, dictionary-based and rule-based methods are gradually eliminated due to the need for manual completion of complex feature, which takes a long time. Statistical machine learning methods are gradually emerging, and their main applications include hidden Markov model[2], maximum entropy model[3], Support vector machines[4]and the conditions random fields[5] etc. However, machine learning methods also have disadvantages such as the need to define features manually and the great dependence on corpus. Last several years, with the enrichment of corpus and the improvement of computing power, deep learning model is widely used in named entity recognition task. Entity recognition is regarded as a task of sequence tagging [6], and gets a good result. 2003 years, Hammerton et al, used LSTM[7] in named entity recognition task for the first time. Collobert et al[8] used the combination of CNN and CRF in the open data set CONLL2003 and had achieved relatively good results. Lample etc[9]used character-level word representation and extracted entities by LSTM and CRF. Huang, etc[10] proposed a BILSTM-CRF model with manually designed spelling features, which reached 88.83% F_1-score in CONLL2003 data set. Chiu and Nichols, etc[11]got the F_1-score of 91.62% in the CONLL2003 data set with the CNN-LSTM model. Li Jianlong et [12] using the BILSTM model, which mixing CNN to process the word vector and integrates the self-attention mechanism, has achieved a good result in entity recognition in the military field. Li Lishuang etc[13]proposed a neural network model based on CNN-BILSTM-CRF, which corpus reaching the best F_1-score of 89.09% and 74.40% in BiocreativeIIGM and JNLPBA 2004 corpus. In order to contain rich semantic information, researchers have successively proposed to use the method of pretrained language model to get the word vector. In October 2018, Google published the BERT model[14], which has refreshed the record of various NLP tasks and received widespread attention and use.

At present, named entity recognition in the general domain has got a good result, but in the specialized domain, due to the different data with different characteristics, it is necessary to adopt effective models for different domains. In the field of health preserving, there is no public data set at present, and the data sources are mainly crawling from mainstream health websites. The data have some unique characteristics. Firstly, the aspects involved in health preserving are relatively complex, so it is necessary to divide the fields and define entity types. This paper divides the health data into six fields: people, diseases, seasons, sports, traditional Chinese medicine and diet. Secondly, due to the different data fields, the data includes both Chinese and some English (such as drug name and disease name). Finally, there are a number of potentially ambiguous words in the data that need attention. For example, "eggs flower" can refer to a kind of herb but can also be a food and they belong to different entity types.

To solve the above problems, BERT-CNN-LSTM-CRF fusion model is proposed on the basis of CNN-LSTM-CRF. The BERT pre-trained language model can transfer learning rich knowledge, and at the same time, compared with Word2vec, this model can effectively distinguish the meanings of polysemous words in different contexts. The proposed model and other models were verified in the self-built health preserving field data set, and it shows that the consequence of the model in this paper is better than other models.

2. Model Description

2.1. Overall framework

The characteristics of the health preserving field are fully considered. The model proposed contains four layers, namely ①BERT layer: BERT is a pre-training model and it uses the superposition of word vector, sentence vector and position vector as input, and Transformer for coding to generate vector representation. ②CNN layer: Further extract local features on the basis of the vector generated by BERT. ③BILSTM layer: further extract global features on the basis of CNN. ④CRF layer: Select the best sequence tagging for BILSTM output. The model fully considers the characteristics of the data, fuses the local and global features and disambiguates, finally outputs the best result. The structure is shown in the Fig. 1.

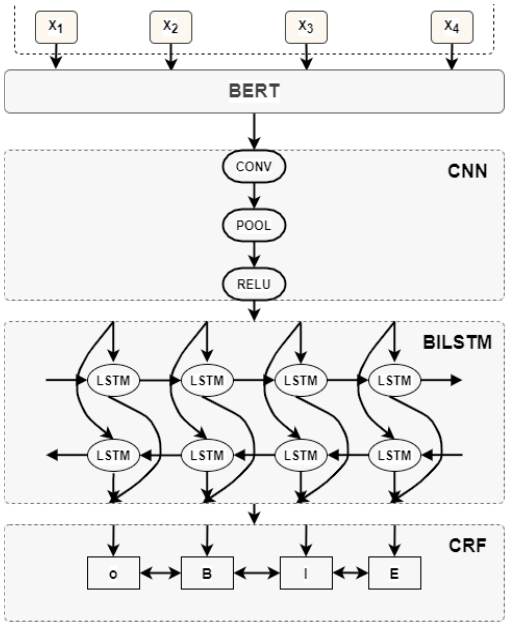

Fig. 1. Structure of Bert-CNN-LSTM-CRF model

2.2. BERT module

In traditional word vector generation methods, such as Word2vec, GloVe and other neural network models, word vectors are mostly context-independent or difficult to learn more contextual information to represent the polysemous nature of words. BERT can fully learn semantic features and generate different vectors for polysemous words according to the scene.

The core of BERT is implemented by multi-layer Transformer encoders, and Transformer is proposed by Vaswani et al[15], which is completely dependent on the self-attention mechanism without using a convolutional neural network or a recursive neural network. The structure of Transformer is shown in Fig.2.

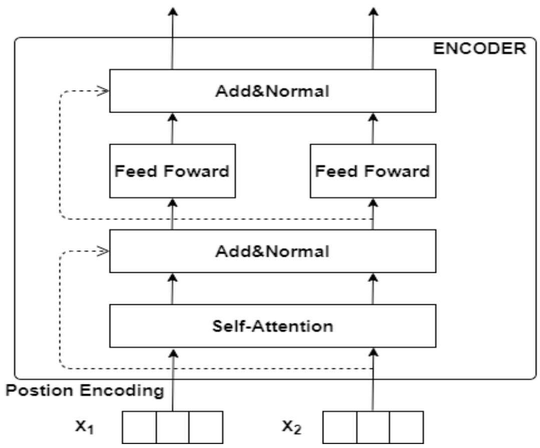

Fig. 2. Transformer structure

Transformer calculates attention for each word in the input sentence with every word in the sentence. The aim is to get the interrelationships between words, capture the sentences' internal structure, and reflect the importance and relevance of different words. The calculation formula is formula (1).

$$Attention(Q,K,V) = soft\max(\frac{QK^T}{\sqrt{d_k}})V \tag{1}$$

In the formula, K, V and Q are Key vector, Value vector and Query vector respectively, d_k is the dimensions of the input vector.

BERT's input vector is generated by overlay of Token embedding, Segment embedding and Position embedding, and adopts two unsupervised tasks, namely, Masked LM and Next Sentence prediction. The former aims at better embedding fusion of context semantics, while the latter aims at considering the connection between adjacent sentences and obtaining the sentence level features.

2.3. CNN module

The convolutional neural network has made great contribution in the field of image classification. Last several years, CNN has also been used to NLP tasks, such as semantic parsing, sentence classification and semantic tagging, and all tasks have achieved good results.

The structure of CNN from left to right are input layer, convolutional layer which is the core, pooling layer and output layer. It uses local connection and weight sharing to carry out convolution operation on input and can well describe local characteristics of data. The convolution process is shown in formula (2).

$$C = f(X \otimes W + b) \tag{2}$$

Where, C represents the output feature map of convolutional layer, and X means the input data, f() means the nonlinear activation function, ⊗ means convolution operation; W means the weight vector of the convolution kernel, and b means the bias term.

The pooling layer can further extract the most representative part of the local features through certain rules, and in the meantime, calculation amount and the quantity of parameters can be reduced to prevent overfitting. It can be expressed by formula (3).

$$P = pool(C) \tag{3}$$

Where, P is output feature map of the pooling layer, pool() is pooling rules, and C means the output feature map of the convolutional layer.

2.4. BILSTM module

Back propagation algorithm is used in deep neural networks, and gradient back propagation is used to guide the updating and optimization of depth network weights according to the error calculated by loss function. With the increase of network depth, gradient disappearance and gradient explosion often occur[16].Therefore, for solving this problem of RNN, LSTM is proposed. LSTM realizes long-term memory through gating concept, and it can also capture sequence information. Its structure can be seen in the Fig.3.

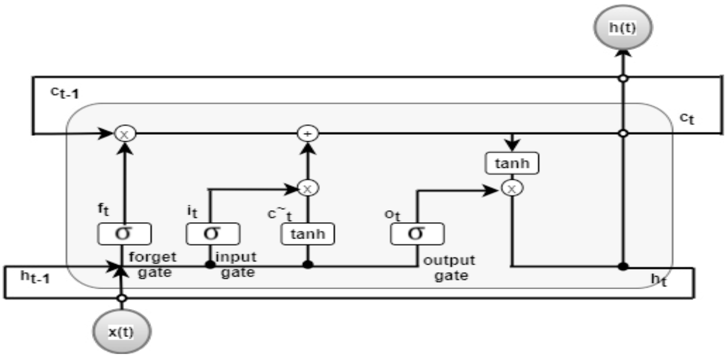

Fig. 3. LSTM logic diagram

Three gate structures are contained in the LSTM: forget gate, input gate and output gate. In this graph: h(t) store short-term status information , c(t) store status information for a long time, when c(t-1) enter time iteration t neurons, it'll first through a forget threshold and lose some memory, then adding some new memories selectively by entering input threshold, c(t) as a long-term state was introduced into the next iteration time t+1 neurons, long-term state to be copied and afferent tanh function at the same time. The results are then filtered by the output threshold, getting short-term state h(t). As shown in formula (4)- formula (8).

$$i_t = \sigma\left(W_{ix}x_t + W_{ih}h_{t-1} + b_i\right) \tag{4}$$

$$f_t = \sigma(W_{fx}x_t + W_{fh}h_{t-1} + b_f) \tag{5}$$

$$o_t = \sigma\left(W_{ox}x_t + W_{oh}h_{t-1} + b_o\right) \tag{6}$$

$$c^{\sim}{}_t = \tanh\left(W_{cx}x_t + W_{ch}h_{t-1} + b_c\right) \tag{7}$$

$$h_t = o_t \otimes \tanh\left(c_t\right) \tag{8}$$

Where, W and b represent the weight matrix and bias vector connecting the two layers respectively; σ represents the sigmoid activation function; ⊗ represents the dot product operation; x_t means the input vector; f_t, o_t and i_t respectively represent the input gate, forget gate and output gate at time t; $\widetilde{c_t}$ is the state at time t, and h_t represents the output at time t.

In order to effectively capture contextual information, BILSTM[17] proposed by Graves A et al will be used. The basic idea of BILSTM is to take forward and backward LSTM for each sequence respectively, and then combine the output at the same time. Therefore, for each moment, it corresponds to forward and backward information, which can effectively obtain global characteristics.

2.5. CRF module

CRF was proposed by Lafferty et al[18] in 2001. It is a discriminative probabilistic graphical model. Linear chain random field is usually used in sequence tagging tasks. For a sequence $x = \{x_1, x_2, x_3 \cdot \cdots x_n\}$, the score of the predictive sequence $y = \{y_1, y_2, y_3 \cdots y_n\}$ can be obtained as:

$$S(x,y)=\sum_{i=0}^{n}A_{y_i,y_{i+1}}+\sum_{i=1}^{n}P_{i,y_i} \tag{9}$$

Where, A represents the transfer matrix, $A_{i,j}$ refers to the tag i transferred to the tag j, P is the output result of BILSTM, and $P_{i,j}$ is the ith word of the jth tag score.

The softmax function is used to get the probability of the sequence y.

$$p(y\,|\,x)=\frac{e^{S(x,y)}}{\sum_{y^{\sim}\in Y_X}S(x,y^{\sim})} \tag{10}$$

Where $\tilde{y}$ is the true tag value, Y_x is all possible of tag sets, and in the training process the likelihood probability of the correct label sequence is maximized:

$$\log\left(P(y\,|\,x)\right)=S(x,y)-\sum_{y^{\sim}\in Y_X}^{n}S(x,y^{\sim}) \tag{11}$$

Finally, Viterbi algorithm is used to get the best predictive tag sequence:

$$y^{*}=\arg\max\left(S(x,y)\right) \tag{12}$$

3. Experiment and analysis

3.1. The experimental data and tagging

The data of health preserving field are all get from health websites. The data contain 8750 sentences after cleaning processing, which are divided into nine entity types based on the data's characteristics, but due to the entities' number of some types are less, so they are put in one class called Others, finally seven classes are got. The date set is divided into training data set, validation set and test set by a 6:2:2 ratio and the distribution of entity categories is shown in the table 1.

Table 1. Number distribution of entity categories.

Entity	Train Set	Validation Set	Test Set
Food	4201	1752	1589
Disease	4128	1586	1224
Drug	1598	601	522
Symptom	3705	1024	1158
Herb	1564	604	558
Benefit	3217	1043	1165
Others	2048	789	691

Commonly used tagging systems for NER including BIO system, BIOE system and BIOES system. BIOES system is selected in this paper, where B is the beginning of an entity, I is the inside of the entity, E is the end of the entity, O is the non-annotated entity, S is the word itself is a separate entity.

3.2. Evaluation indexes and experimental parameters

In this paper, three indexes including precision, recall and F_1-score are adopted to evaluate the named entity recognition effect. Their calculation methods are as follows:

$$P = \frac{TP}{TP + FP} \tag{13}$$

$$R = \frac{TP}{TP + FN} \tag{14}$$

$$F_1 = \frac{2PR}{P + R} \tag{15}$$

Where, TP is the number of correctly identified entities. FP is the number of entities identified with errors, FN is the number of unrecognized entities.

In this experiment, Python version 3.7 and TensorFlow 1.14.0 are adopted, BERT version is Bert-base-Chinese, which has 12 layers of Transformer and 12 heads of attention mechanism, the dimension of the hidden layer is 768. Adam optimizer is used, learning rate is 0.001, and dropout is 0.5, batch_size is 64, and epoch is 50.

3.3. Experimental results and analysis

In order to verify the performance of this model in the field of health-preserving named entity recognition, BILSTM-CRF, CNN-LSTM-CRF and LSTM-CRF are selected to conduct comparative experiments on the same data set, and the result is shown in Table 2.

Table 2. Experimental result of different models (%)

Model	P	R	F_1
LSTM-CRF	82.56	79.01	80.75
BILSTM-CRF	84.13	82.28	83.19
CNN-BILSTM-CRF	86.48	84.47	85.46
BERT-CNN-BILSTM-CRF	89.54	86.21	87.84

In order to verify the effectiveness of BILSTM, comparative experiments of LSTM-CRF and BILSTM-CRF were carried out, and the results showed that the use of BILSTM increased by 2% to 3% in P, R and F_1. Compared with LSTM, BILSTM could take full advantage of contextual information.

To verify the effectiveness of CNN, BILSTM-CRF and CNN-BILSTM-CRF were compared, and the results showed that the precision increased by 2.35%, the recall increased by 2.1%, and the F_1-score increased by 2.27%, indicating the effectiveness of CNN in extracting local features, and the recognition effect of CNN in including Chinese and English mixed characters and special characters was very good.

To verify the effectiveness of the BERT pre-training model, has carried on the CNN-BILSTM-CRF and BERT-CNN-BILSTM-CRF contrast experiment, the results show that the three indexes have a lot to improve, achieving the best F_1-score of 87.84%. It suggests that BERT can find good vector in flexible expression and the ambiguity of words and improve the ability of feature extraction.

Besides, as shown in the Fig.4, we also analyze the changes of the F_1-score of each model with the training process. It can be seen that BERT pre-training language model can reach a high level at the initial stage of training and then continue to rise and soon become stable. In this process, the F_1-score of BERT pre-training language model is always better than that of other models. The other models have a low initial value and need several rounds of iteration to reach a high level. After 50 rounds of iteration, each model tends to be stable, and the F_1-score of this model proposed in this paper is the best.

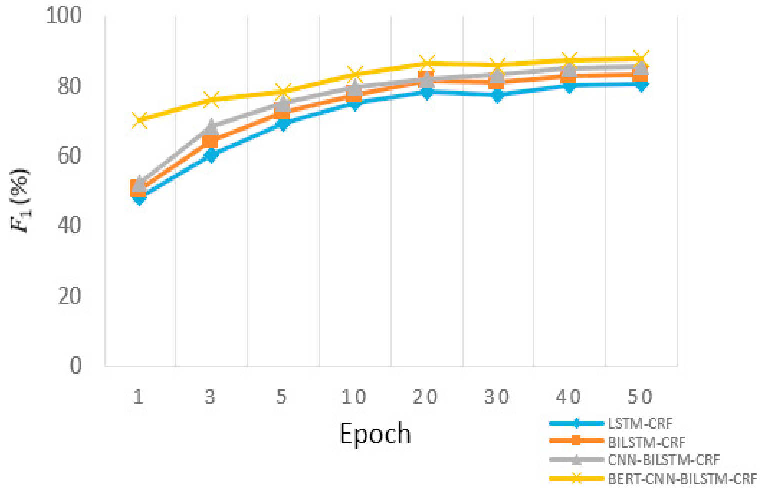

Fig. 4. The F_1-score update diagram

4. The conclusion

NER task is the foundation of knowledge graph. For named entity recognition task in the field of health preserving, in this paper, through crawling data from websites to establish the data set in the field of health preserving, defining the seven types of entities, and proposing a model of named entity recognition based on BERT. BERT can make full use of the context semantic to address the problem of words ambiguity and generate different vectors, then use the local feature extraction, CNN is used to the local feature extraction and BILSTM is used to identified the global characteristics, finally by CRF choosing the best sequence tagging. With BILSTM-CRF, CNN BILSTM-CRF and LSTM-CRF models experiment on the same data set, verifying the advantages of the model proposed in this paper, achieving the best F_1-score, 87.84%. This model provides the foundation for building knowledge graph and completing the question answering system. In the follow-up research, we will continue to expand data sets, and consider how to simplified the model under the premise of no reducing the recognition effect.

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References

1. Liu Liu, Wang Dong Bo. Summary of Research on Named Entity Recognition[J]. Acta Information, 2008,37(3): 329-340
2. Yu Hong Kui, Zhang Hu Ping, Liu Qun, et al. Chinese Named Entity Recognition Based on Cascading Hidden Markov Model [J]. Journal on Communications, 2006,27 (2): 87-94.
3. McCallum A, Freitag D, Seventeenth International Conference on Machine Learning. Sydney, 2000: 171-598. Seventeenth International Conference on Machine Learning: Seventeenth International Conference for Information Extraction and segmentation [C]
4. Ertekin S, Bottou L. Nononline support vector machines [J]. Transactions on Pattern Analysis and Machine Intelligence, 2011,33 (2) :368-381
5. Zhou Junsheng, Dai Xinyu, Yin Cunyan, et al. Automatic Recognition of Chinese Organization Name based on Cascading Random Field Model [J]. Acta Electronica Sinica, 2006,34 (5) :804-809.

6. Wang Lei, Xie Yun, Zhou Junsheng, Gu Yanhui, Qu Weiguang. Seging-level Chinese Named Entity Recognition Based on Neural Network [J]. Journal of Chinese Information Processing, 2016,32(3):84-90.
7. Hammerton J. Named Entity Recognition with Long Short Term Memory[C]// Seventh Conference on Natural Language Learning at Hlt-naacl. Association for Computational Linguistics, 2003.
8. Collobert R, Weston J, Bottou, Leon, et al. Natural Language Processing (Almost) from Scratch[J]. Journal of Machine Learning Research, 2011, 12(1):2493-2537.
9. Lample G, Ballesteros M, Subramanian S, et al. Neural Architectures for Named Entity Recognition[J].2016.
10. Huang Z, Xu W, Yu K. Bidirectional LSTM-CRF Models for Sequence Tagging[J]. Computer Science,2015.
11. Chiu J P C, Nichols E. Named Entity Recognition with Bidirectional LSTM-CNNS [J]. ArXiv Preprint arXiv:1511.08308,2015.
12. Li Jianlong, Wang Panqing, Han Qiyu. Recognition of military named entities based on BILSTM [J]. Computer Engineering and Science,2019,41(4):713-718.
13. Li Lishuang, Guo Yuankai. Biomedical Name Entity Recognition with CNN-BLSTM-CRF[J]. Journal of Chinese Information Processing, 2012,32 (1) :117122.
14. Devlin J, Chang M W, Lee K, et al. BERT: pre-training of deep bidirectional transformers for language understanding[C]//Proceedings of the 2019 Conferenee of the Association for Computational Linguistics. Stroudsburg,2019:4171-4186.
15. Vaswani A, Shazeer N, Parmar N, et al. Attention is all you need [C] / /Proceedings of the 31st International Conference on Neural Information Processing Systems. New York, 2017: 6000-6010.
16. Bengio Y, Lamblin P, Popovici D, et al. Greedy Layer - Wise Training of deep Network. Advances in Neural Information Processing System, 2007,6 (19) :153-162.
17. Graves A, Framehuber J. Framework wise Phoneme Classification with Bidirectional LSTM and other Neural Network architectures. Neural Network, 2005, 18(5-6): 602-610.
18. Lafferty, John D, McCallum, et al. Conditional Random Fields: Probabilistic models for Segmenting Sequence Data [C] /Proc of the 18th International Conference on Machine Learning. 2001:282-289.